



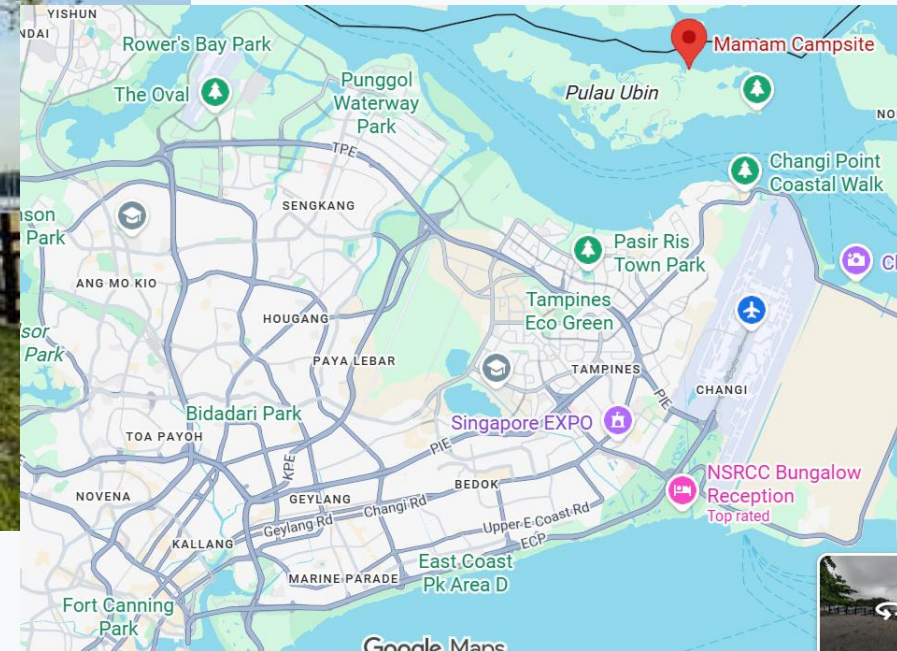
Case Study: Solving Problems with Quantum Annealing

QUBO formulation of the Number Partitioning Problem and sampling
with Quantum Annealing

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Everyday Example: Splitting Luggage



Everyday Example: Splitting Luggage

We are going camping at Mamam Campsite.
We want to split the luggage evenly between two cars.

Tent 3 kg

Sleeping bag 1 kg

Food 4 kg

Cookware 2 kg

Water 2 kg

Total: 12 kg → ideal split: 6 kg / 6 kg

This is an optimization problem called the Number Partitioning Problem (NPP).

NPP: Split numbers into two groups so that the difference between their sums is small.

Everyday Example: Splitting Luggage

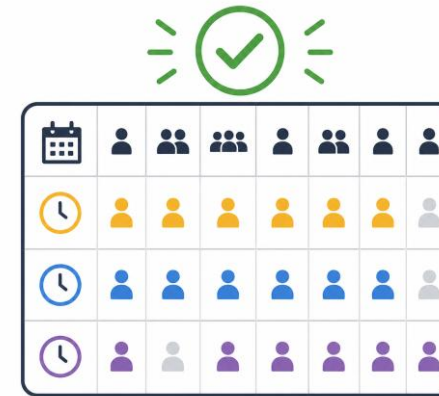
There are many optimization problems other than NPP.



Portfolio
optimization



Traffic-flow
optimization



Shift
scheduling

etc...

Quantum Annealing (QA) is expected to be a useful method for solving such problems efficiently.

QUBO: A binary quadratic optimization form used in QA

The target energy is represented by 0/1 variables and a quadratic interaction matrix.

QUBO form

$$H(\mathbf{x}) = \sum_i Q_{ii} x_i + \sum_{i < j} Q_{ij} x_i x_j$$

$x_i \in \{0,1\}$

- **Quadratic**: includes quadratic terms $x_i x_j$
- **Unconstrained**: constraints are added as penalty terms, not handled directly
- **Binary**: each variable is 0 or 1
- **Optimization**: find the variable assignment that minimizes $H(\mathbf{x})$

Image of the interaction matrix Q

	X ₀	X ₁	X ₂	X ₃	X ₄
X ₀	Q ₀₀	Q ₀₁	Q ₀₂	Q ₀₃	Q ₀₄
X ₁		Q ₁₁	Q ₁₂	Q ₁₃	Q ₁₄
X ₂			Q ₂₂	Q ₂₃	Q ₂₄
X ₃				Q ₃₃	Q ₃₄
X ₄					Q ₄₄

Because $x_i^2 = x_i$ for binary variables, linear terms can also be handled.

In QA, we search for x that makes $H(x)$ small.

Representing NPP with 0/1 variables

To formulate NPP as a QUBO, represent each item with a 0/1 variable.

Items and weights

i	Item	a_i
0	Tent	3
1	Sleeping bag	1
2	Food	4
3	Cookware	2
4	Water	2

Weight vector $a = [3, 1, 4, 2, 2]$

Meaning of the variables

$x_i = 1 \rightarrow$ item i goes to Car A

$x_i = 0 \rightarrow$ item i goes to Car B

This variable assignment is a candidate solution.

Example: $x = [1, 0, 1, 0, 0]$

Defining the objective function

Objective function

$$H(\mathbf{x}) = \left(\sum_{i=1}^N a_i (2x_i - 1) \right)^2$$

↓

This means “sum of A – sum of B”

$$\begin{aligned} x_i = 1 &: a_i(2x_i - 1) = a_i \\ x_i = 0 &: a_i(2x_i - 1) = -a_i \end{aligned}$$

- $H(\mathbf{x})$ becomes larger when the two sums are more different
- Squaring gives a quadratic expression that can be expanded into a QUBO

Let $A = \sum_i a_i$,

$$\begin{aligned} H(\mathbf{x}) &= \left(2 \sum_i a_i x_i - A \right)^2 \\ &= \sum_i \underbrace{(4a_i^2 - 4Aa_i)}_{Q_{ii}} x_i + \sum_{i < j} \underbrace{8a_i a_j}_{Q_{ij}} x_i x_j + A^2 \end{aligned}$$

The objective function is in QUBO form

$$H(\mathbf{x}) = \sum_i Q_{ii} x_i + \sum_{i < j} Q_{ij} x_i x_j$$

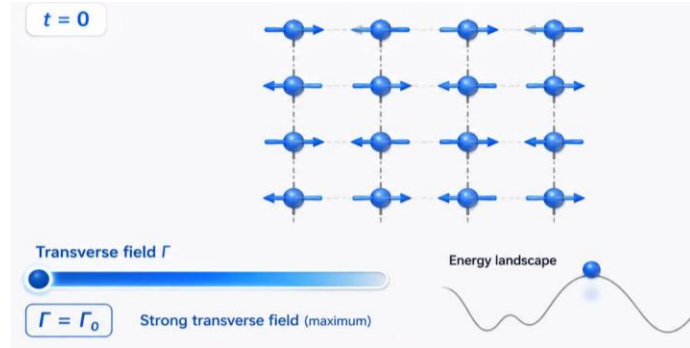
We minimize this $H(x)$ with QA.

QA: Reduce the transverse field to find the ground state

Add a transverse-field term H_T to the objective function H .

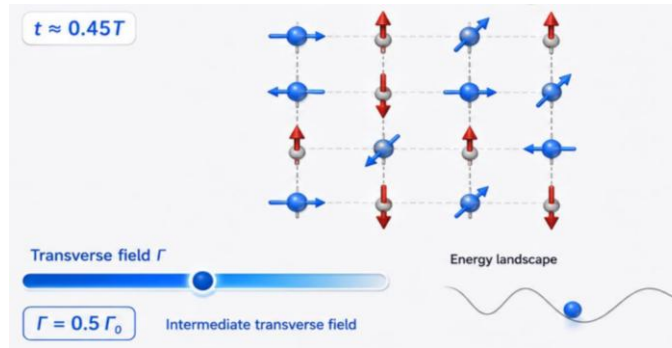
$$H_{QA}(t) = A(t)H_T + B(t)H$$

Start with broad exploration, then gradually approach a low-energy state of the problem Hamiltonian.



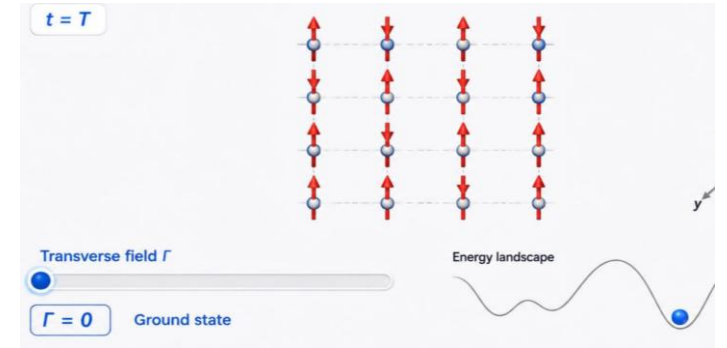
① Strong transverse field

Explore a wide range of states



② Reduce the transverse field

Follow the energy landscape of the problem



③ Measure the ground state

Obtain a low-energy solution (variable assignment)

Strong quantum fluctuation $\rightarrow$ reduce fluctuation $\rightarrow$ read out a low-energy state

Demo: Run the program in Colab

Please open the file named `QI4U\_Case\_Study\_code.ipynb` from the drive attached to Discord or a similar platform.

Interpreting sampled variables in the original problem

Example of the best variable assignment

$$x = [0, 0, 1, 0, 1]$$

1 → Car A / 0 → Car B

Car A

Food 4 kg

Water 2 kg

Total: 6 kg

Car B

Tent 3 kg

Sleeping bag 1 kg

Cookware 2 kg

Total: 6 kg

Difference = 0 kg → $H(x)=0^2=0$

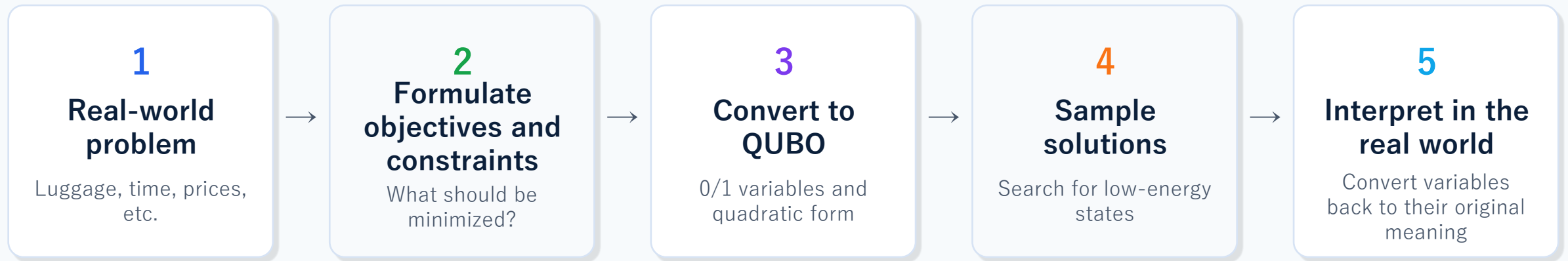
Items and weights

i	Item	a_i
0	Tent	3
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4	Water	2

A bitwise-inverted optimal solution represents the same split, so multiple good solutions may appear.

By checking the frequency of samples, we can see solution stability and diversity.

Summary: How to solve a problem with QA



Summary: How to view problem solving with QA

Key points of this material

QA treats the objective function as energy and searches for low-energy states.
By formulating a problem as a QUBO, we can solve optimization problems with QA.

QUBO represents a problem using 0/1 variables and a quadratic form.
For NPP, the objective function is the squared difference between the two group sums.

QA samples solutions multiple times.
Sampling lets us observe several good solution patterns.